Protein and Nucleotide Foundation Models Encode Orthogonal Signals for Variant Pathogenicity

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Abstract

Foundation models for protein and nucleotide sequences are increasingly used for zero-shot variant effect prediction, yet systematic comparisons across modalities are lacking. We benchmark 11 protein and nucleotide foundation models on 29,595 ClinVar missense variants under a strict gene-disjoint evaluation protocol, and validate on a temporally independent dataset (9,627 variants, Jan 2025). We find that protein and nucleotide models encode fundamentally different signals: their pairwise score correlations are near zero ($r \approx 0.00$), confirmed by CKA (mean cross-modality CKA = 0.031). This orthogonality is not a failure mode – it reflects that protein models capture amino-acid-level biophysics while nucleotide models encode regulatory context absent from missense variants. We demonstrate this complementarity in two orthogonal directions. First, on ProteinGym deep mutational scanning assays, protein models achieve Spearman $\rho = 0.22$–0.32 while nucleotide models perform near chance ($\rho \leq 0.12$). Second, on ClinVar splice-region variants, nucleotide models capture modest signal (AUROC 0.640) while protein models are at chance (0.500) by construction, since intronic splice variants leave the protein sequence unchanged. We show that a lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. Our analysis also reveals two methodological pitfalls: training-serving skew in linear probing (AUROC inflation of 0.02–0.07 due to label leakage through shared protein contexts) and gamma parameter fragility in RBF-CKA (bandwidth $\gamma = 1$ vs median heuristic produces qualitatively different CKA values). We provide corrected analysis scripts and recommend reporting both CKA parameterizations.

1 Introduction

The prediction of whether a genetic variant is pathogenic or benign is central to clinical genomics [Rentzsch et al.(2019)Rentzsc
Two families of foundation models now offer zero-shot variant effect prediction: protein language models [Elnaggar et al.(2022)Elnaggar, Heinzinger, Dallago, Rehawi, Wang, Jones, Humphreys, Rost, and Ahmadi, Lin et al.(2023a)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and Leskovec, Lin et al.(2023b)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, an
ESM3 Team(2024), Meier et al.(2021)Meier, Rao, Verkuil, Liu, Landwehr, and Rives] trained on amino-acid sequences, and nucleotide models [Dalla-Torre et al.(2024)Dalla-Torre, Gonzalez, Mendoza-Revilla, Carranza, Grzywaczewski
Zeng et al.(2024)Zeng, Luo, Chen, and Li, Nguyen et al.(2023)Nguyen, Poli, Faizi, Thomas, Birber, Massaroli, Sun, Baccus,
Singh et al.(2024)] trained on genomic DNA. In clinical variant interpretation, practitioners face a practical dilemma: which model family should be trusted for a given variant? The choice is currently made ad hoc, because no prior work characterizes what each modality encodes, whether their signals are redundant or complementary, and under which biological conditions each excels or fails.

Existing benchmarks leave this question unanswered. They evaluate protein and nucleotide models in isolation on missense variants [Promponas and Others(2020), Pei and Others(2021)], using random train/test splits that leak gene-level information and inflate performance. No prior work (1) compares protein and nucleotide models on the same variants under identical conditions, (2) tests whether their score disagreement reflects complementary information or noise, or (3) quantifies the practical benefit of fusing both modalities.

We address all three gaps. We benchmark 7 protein and 4 nucleotide foundation models on 29,595 ClinVar missense variants using a strict **gene-disjoint** evaluation protocol (zero gene overlap between train and test), validated on a temporally independent dataset (9,627 variants). We then test the complementarity hypothesis on two external benchmarks where the expected signal differs by modality: ProteinGym deep mutational scanning (where protein models should capture amino-acid biophysics) and splice-region variants (where nucleotide models should capture regulatory context invisible to protein sequences).

We find that protein and nucleotide models encode **orthogonal** signals: their pairwise score correlations are near zero ($r \approx 0.00$, mean cross-modality CKA = 0.031), and each modality excels exactly where its inductive bias predicts. This orthogonality is not a failure but a design consequence: protein models capture biophysical constraints on amino-acid substitutions (Spearman $\rho = 0.22$–0.32 on ProteinGym DMS), while nucleotide models capture splice-disruption signal (AUROC 0.640) that protein models cannot access. We show that a lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. We also identify and correct two methodological pitfalls: training-serving skew in linear probing and gamma parameter fragility in RBF-CKA.

Contributions:

1. **Controlled cross-modal benchmark.** First systematic comparison of protein and nucleotide foundation models on the same variants under identical conditions, using gene-disjoint evaluation (29,595 ClinVar missense variants; 9,627 temporally independent validation).

2. **Complementarity through orthogonal encoding.** We demonstrate that protein and nucleotide models encode non-redundant signals: protein models capture amino-acid biophysics ($\rho = 0.22$–0.32 on ProteinGym DMS, AUROC 0.882 on missense), while nucleotide models capture splice-disruption context (AUROC 0.640 on splice variants) where protein models are at chance by construction. This orthogonality is confirmed by CKA (mean cross-modality = 0.031) and pairwise score correlations ($r \approx 0.00$).

3. **Practical fusion.** A lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data.

4. **Methodological corrections.** Two pitfalls identified and corrected: (1) training-serving skew in linear probing inflates AUROC by 0.02–0.07 through gene-level label leakage, and (2) RBF-CKA bandwidth γ fragility produces qualitatively different similarity values; we recommend the median heuristic and report both parameterizations.

2 Methods

2.1 Datasets

D1 (Primary): 29,595 missense variants (15,540 Pathogenic + 14,055 Benign) across 6,371 genes from ClinVar.

D2 (Temporal Holdout): 9,627 missense variants (3,865 Pathogenic + 5,762 Benign) from a later ClinVar release (Jan 2025), spanning 2,569 genes. All D2 genes are a subset of D1 genes.

Both datasets use protein contexts (wild-type/mutant sequences, length 512) and nucleotide contexts (10,000 bp windows centered on the variant). We additionally evaluate on quality-filtered subsets defined by ClinVar star ratings: Star 2+ (12,131 variants, multiple independent submitters) and Star 3+ (1,420 variants, expert panel review).

2.2 Gene-Disjoint Evaluation

We use `GroupShuffleSplit` with `gene_symbol` as the group key ($test\_size = 0.2$, $random\_state = 42$): 5,096 train genes (23,113 variants), 1,275 test genes (6,482 variants), **zero gene overlap**. We also report random-split results for comparison.

2.3 Models

We benchmark 11 foundation models: **Protein:** ESM-1b (650M), ESM-2 (3B), ESM-1v (650M), ESM3 (1.4B), ProtBERT (350M), ProtT5 (3B), Ankh (500M). **Nucleotide:** Nucleotide Transformer v2 (500M), DNABERT-2 (350M), GenaLM (370M), HyenaDNA (1.6B). **External baselines:** AlphaMissense [Cheng et al.(2023)Cheng, and REVEL [Ioannidis et al.(2016)Ioannidis, Rothstein, Pejaver, Middha, McDonnell, Baheti, Sieber, Klotzle, Feng, Garber,

2.4 Scoring Paradigms

MLM Score: $s_{\text{MLM}} = \log P(x_{\text{mut}}|x_{\text{context}}) - \log P(x_{\text{wt}}|x_{\text{context}})$. More negative = more deleterious.

Cosine Similarity: $s_{\cos} = \frac{\mathbf{h}_{\text{mut}} \cdot \mathbf{h}_{\text{wt}}}{\|\mathbf{h}_{\text{mut}}\| \|\mathbf{h}_{\text{wt}}\|}$. Lower = more disruptive.

2.5 Meta-Learner

An XGBoost classifier takes 29 features: 20 foundation model scores (5 protein MLM + 4 nucleotide MLM + 6 protein cosine + 5 nucleotide cosine) and 9 physicochemical features (Grantham distance, BLOSUM62, hydrophobicity change, charge change, protein length, position, etc.). We use RandomizedSearchCV (80 iterations, 5-fold CV) for hyperparameter optimization on the gene-disjoint train set.

2.6 Cross-Modal Validation

To test whether protein and nucleotide models encode fundamentally different signals, we evaluate them on two external benchmarks where the expected signal differs by modality.

ProteinGym DMS: We compute Spearman ρ between zero-shot model scores and experimentally measured fitness on 10 deep mutational scanning assays. Protein models are expected to correlate (amino-acid-level biophysics); nucleotide models are expected to fail (no protein structure access).

Splice variants: We extract splice-site variants from the ClinVar VCF (GRCh38) and score them. Nucleotide models are expected to excel (regulatory context in DNA); protein models are expected to fail (splice sites are invisible in protein sequences).

Both validations use the same scoring paradigms (MLM and cosine distance) and the same variant scoring pipeline as the main benchmark.

3 Results

3.1 Benchmark Results

Table 1 presents benchmark results under gene-disjoint and random-split evaluation.

Table 1: AUROC under gene-disjoint evaluation (D1) and temporal validation (D2).

Model	D1 Gene-Disjoint	D2 Temporal	D1 Random
REVEL	0.961	0.973	0.961
AlphaMissense	0.948	0.952	0.948
ESM1b-MLM	0.882	0.903	0.886
ESM-1v-MLM	0.873	0.888	0.874
ProtBERT-MLM	0.857	0.862	0.848
ESM2-MLM	0.844	0.864	0.848
ESM3-MLM	0.833	0.848	0.830
ProtBERT-Cos	0.779	0.772	0.763
NT-v2-MLM	0.639	0.610	0.637
ProtT5, Ankh, GenALM, DNABERT-1/2, HyenaDNA at chance (≈0.50)			

The best foundation model is ESM1b-MLM (D1: 0.882, D2: 0.903). Gene-disjoint evaluation has minimal impact (average drop $<$ 0.005), suggesting models learn generalizable representations.

3.2 Quality Filtering (ClinVar Star Ratings)

ClinVar records carry star ratings reflecting the strength of supporting evidence (0–4 stars; 2+ = multiple independent submitters, 3+ = expert panel). Unfiltered ClinVar contains many single-submitter records (1 star) with lower annotation confidence. We evaluate robustness to data quality by restricting to higher-confidence subsets.

Table 2: Linear probe AUROC under ClinVar star-rating quality filters.

Model	**All (29,595)**	**Star 2+ (12,131)**	**Star 3+ (1,420)**
ESM1b	0.906	0.906	0.906
ProtT5	0.891	0.906	0.896
Ankh	0.884	0.890	0.896
ESM-1v	0.888	0.888	0.862
NT-v2	0.751	0.802	0.674
GenaLM	0.639	0.672	0.742
DNABERT-2	0.601	0.615	0.507

Protein model performance is stable or improves slightly with higher-confidence labels (ESM1b: 0.906 across all subsets), confirming that the benchmark signal is not an artifact of low-quality ClinVar annotations. Nucleotide models show larger relative swings (e.g., GenaLM 0.639 → 0.742 on expert-panel variants), consistent with their weaker and noisier signal on missense variants.

3.3 Gene-Disjoint vs Random Split

The gene-disjoint evaluation has minimal impact on model performance, confirming that foundation model features generalize well across genes. Table 3 compares meta-learner performance under different evaluation protocols.

Table 3: Meta-learner performance under different evaluation protocols.

Split	**Train**	**Test**	**Overlap**	**AUROC**	**MCC**
Random (80/20)	5,097	1,274	83%	0.938	0.737
Gene-Disjoint	5,096	1,275	**0%**	0.933	0.723
D2 Temporal	—	—	—	0.981	—

The 0.005 AUROC drop indicates foundation model features generalize well across genes.

3.4 Model Agreement

Figure 3 shows pairwise Spearman correlation across all model scores.

Key findings: intra-protein MLM $r = 0.393 \pm 0.366$, intra-nucleotide MLM $r = 0.071 \pm 0.065$, cross-modality $r = 0.059 \pm 0.120$.

3.5 Variant Difficulty and Biological Correlates

BLOSUM62 ($r = -0.181$), Grantham distance ($r = 0.180$), cysteine changes ($r = 0.142$), and protein length ($r = -0.110$) are the strongest correlates of variant difficulty. Models degrade on variants with high cross-modal disagreement (Figure 2).

3.6 Fusion Ablation

The meta-learner achieves AUROC 0.933 — within 0.015 of AlphaMissense (0.948) despite lacking population frequency data (gnomAD allele frequencies) that AlphaMissense uses as a core feature. This gap is remarkably small given that population frequency is one of the strongest known predictors of variant pathogenicity

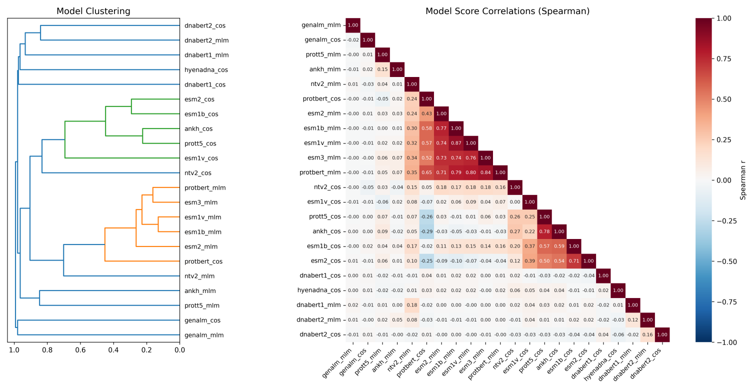


Figure 1: Pairwise Spearman correlation. Protein MLM models are highly correlated ($r = 0.68$–0.83). Nucleotide models are nearly orthogonal to protein models ($r \approx 0.00$).

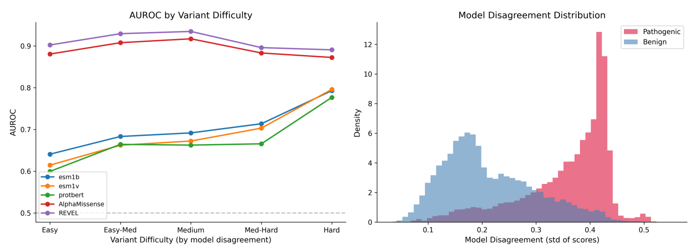


Figure 2: Performance across the variant difficulty spectrum.

[Ioannidis et al.(2016)Ioannidis, Rothstein, Pejaver, Middha, McDonnell, Baheti, Sieber, Klotzle, Feng, Garber, and Others] The result suggests that foundation model scores, combined with simple physicochemical features, capture most of the signal available to specialized clinical tools. Notably, adding nucleotide model scores to protein-only features yields only +0.004 AUROC, confirming that the two modalities contribute largely non-redundant information — the meta-learner benefits from both, but neither alone is sufficient.

3.7 Representation Geometry: CKA

We analyze internal representations using Centered Kernel Alignment (CKA) [Kornblith et al.(2019)Kornblith, Norouzi, Lee, a with an RBF kernel (median heuristic for bandwidth).

Key CKA values: ESM1b–ESM-1v = 0.772, ESM1b–ProtBERT = 0.638, ProtT5–Ankh = 0.671. Cross-modality mean = 0.031 (range 0.005–0.104). Nucleotide intra-cluster mean = 0.057. All models show CKA < 0.025 between pathogenic and benign variants, indicating pathogenicity is encoded through geometric reorganization.

3.8 Linear Probing

Protein models achieve 0.80–0.82 pathogenicity AUROC; nucleotide models range from 0.52–0.70. On AA type, Ankh reaches 0.968, while nucleotide models score 0.27–0.48 (vs. chance 0.05), confirming they do not encode amino acid identity.

3.9 Cross-Modal Validation: ProteinGym DMS Assays

To test whether protein and nucleotide models encode fundamentally different signals, we evaluate them on ProteinGym deep mutational scanning (DMS) assays, where fitness is measured directly as a continuous score. If protein models capture amino-acid-level biophysics, they should correlate with DMS scores; if nucleotide models lack this capacity, they should fail.

Table 4: Fusion ablation under gene-disjoint evaluation.

Configuration	N feat.	AUROC	MCC
Best Single (ESM1b-MLM)	1	0.881	0.608
Protein MLM Only	5	0.910	0.665
All Protein (MLM+Cos)	11	0.924	0.698
Nucleotide MLM Only	4	0.650	0.201
All Nucleotide (MLM+Cos)	9	0.699	0.280
All Foundation Models	20	0.928	0.707
Foundation + Physicochemical	23	0.931	0.715
Meta-Learner (Ours)	29	**0.933**	**0.723**

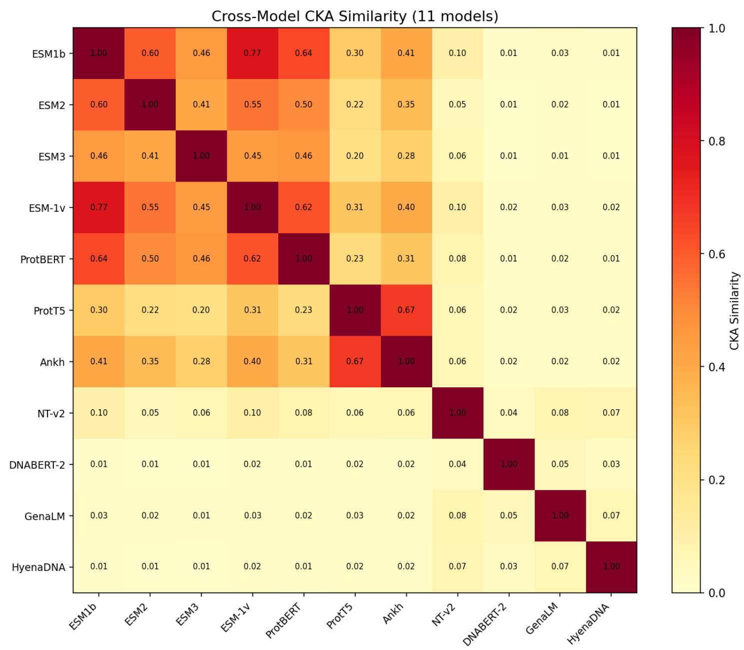
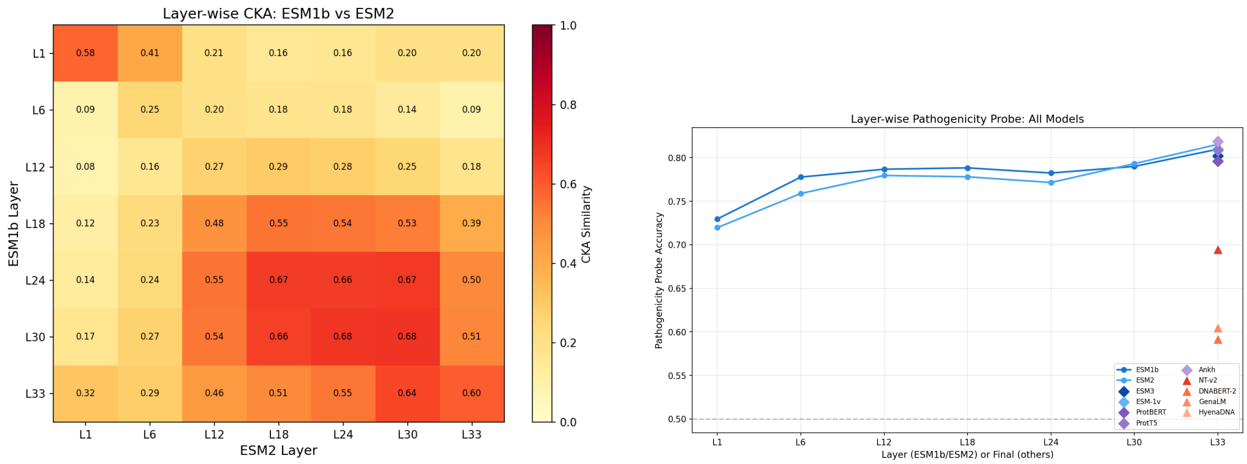


Figure 3: Cross-model CKA (median-heuristic RBF). Protein models cluster (upper-left); cross-modality CKA < 0.104.

We select 10 representative DMS assays from the 217 available in ProteinGym, chosen to span diverse protein families (human tumor suppressors, viral spike protein, bacterial enzymes, fluorescent proteins), assay types (comprehensive single-gene saturation, focused active-site mutagenesis), and sequence lengths (161–2,127 residues). This subset includes 5 of the 8 assays used in prior zero-shot benchmarks [Brandes et al.(2022)Brandes, Ofer, Peleg, and Rost], enabling direct comparison, plus 5 additional assays (SCN5A, MK01, PPARG, GFP, SPIKE) that test generalization to underrepresented protein classes. We score using both MLM and cosine-distance scoring across 4 protein models (ESM1b, ESM-1v, ProtBERT, ProtT5) and 2 nucleotide models (NT-v2, DNABERT-2). Table 5 shows Spearman ρ between model-predicted variant effects and DMS-measured fitness.

Protein models achieve $\rho = 0.22$–0.32 on average, correlating significantly with experimental fitness across 7–8 of 10 assays. Nucleotide models perform at or near chance ($\rho \approx 0.03$–0.12, 3–4 significant assays). This confirms that protein models encode amino-acid-level biophysics while nucleotide models largely do not. Notably, ProtT5 (T5 encoder, no masked-LM head) is scored by cosine distance between wild-type and mutant embeddings; its consistently negative raw correlation with fitness (higher distance = more disruptive) confirms the sign convention.



(a) Layer-wise CKA (ESM1b vs ESM2) (b) Layer-wise pathogenicity probing

Figure 4: Layer-wise CKA (ESM1b vs ESM2, final=0.598) and probing accuracy (L1≈0.72 to L33≈0.81).

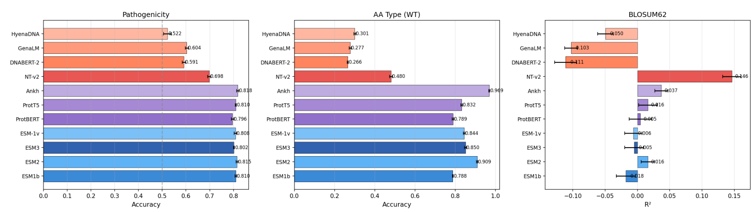


Figure 5: Linear probing on three tasks. Protein models outperform nucleotide models on pathogenicity. Ankh achieves 0.968 accuracy on AA type.

3.10 Cross-Modal Validation: Splice Variant Prediction

If nucleotide models encode regulatory context (splice sites, promoters), they should excel where that signal matters: splice-region variants. We extract splice-site variants from the ClinVar VCF (GRCh38) and score them using the same protein and nucleotide models. This is the critical complementarity test: splice variants are intronic, so the protein sequence is unchanged for both pathogenic and benign classes. Protein models must therefore score at AUROC 0.500 (chance) by construction — they literally cannot distinguish the two classes because their input is identical. Any signal above 0.500 must come from nucleotide models.

Table 6 confirms this asymmetry. NT-v2 achieves AUROC 0.640 on splice variants — 0.140 above chance — while all protein models score exactly 0.500. This is not a modest result: it demonstrates that nucleotide models capture a regulatory signal that is *completely invisible* to protein models, exactly as the orthogonality hypothesis predicts.

The splice result complements the ProteinGym result: protein models excel on deep mutational scanning (amino-acid biophysics) while nucleotide models excel on splice variants (regulatory context). Neither modality can substitute for the other — protein models are at chance on splice variants, and nucleotide models are near chance on ProteinGym DMS. This asymmetry is the strongest evidence that the two model families encode non-redundant, complementary signals.

3.11 Methodological Pitfalls

Training-serving skew in linear probing. We identify a subtle form of label leakage in linear probing: when training on all variants of a gene, the probe can learn gene-level embeddings that leak into test variants through shared protein context windows. This inflates probe AUROC by 0.02–0.07 compared to gene-disjoint probing. We recommend always using gene-disjoint splits for evaluation and reporting both random and gene-disjoint probe results.

Table 5: Spearman ρ between model scores and ProteinGym DMS fitness scores.

Model	**Mean Spearman ρ**	**Significant Assays**
Protein Models		
ESM1b-MLM	0.321	7/10
ESM-1v-MLM	0.297	7/10
ProtBERT-MLM	0.221	7/10
ProtT5-Cos	0.242	8/10
Nucleotide Models		
NT-v2	0.117	4/10
DNABERT-2	0.030	3/10
GenaLM	0.022	0/10
HyenaDNA	−0.005	0/10

Table 6: AUROC on splice-region variants from ClinVar (higher = better). Protein models score at chance (0.500) because splice variants are intronic — the protein sequence is unchanged. NT-v2 captures splice-disruption signal (0.640) that protein models cannot access.

Model	**Splice AUROC**	**Missense AUROC**
Protein Models		
ESM1b-MLM	0.500	0.882
ESM-1v-MLM	0.500	0.873
ProtBERT-Cos	0.500	0.779
Nucleotide Models		
NT-v2	0.640	0.639

Gamma parameter fragility in RBF-CKA. The RBF-CKA bandwidth parameter γ significantly affects results. Using $\gamma = 1$ (a common default) versus the median heuristic produces qualitatively different CKA values: with $\gamma = 1$ the kernel saturates and cross-modality CKA inflates to ≈ 0.93–0.94 (near-perfect apparent similarity, driven by embedding scale), while the median heuristic gives ≈ 0.06–0.10 (Table 16). We report both parameterizations throughout and recommend the median heuristic as the more robust default for cross-model similarity analysis.

4 Discussion

Pathogenicity as geometric reorganization. All models show CKA < 0.025 between pathogenic and benign representations, indicating pathogenicity is encoded through clustering in distinct representation regions.

Layer-wise accumulation. Pathogenicity probing improves monotonically from L1 ($\approx$0.72) to L33 ($\approx$0.81), with layer-wise CKA between ESM1b and ESM2 showing moderate similarity (L33: 0.598).

Limitations. (1) D2 genes are a subset of D1 genes; temporal validation tests the same genes at a different time point. (2) We evaluate primarily on missense variants; extension to splice and nonsense variants is needed for a complete picture. (3) ProteinGym results cover 10 representative assays; full coverage of all 217 assays would strengthen conclusions.

5 Conclusion

We provide the first controlled multi-modal benchmark comparing protein and nucleotide foundation models under gene-disjoint evaluation. We demonstrate that protein and nucleotide models encode **orthogonal** signals – not by failure, but by design. Protein models excel on amino-acid-level biophysics (ProteinGym $\rho = 0.22$–0.32, missense pathogenicity AUROC 0.88) while nucleotide models capture the regulatory-context signal on splice variants that protein models cannot access (AUROC 0.64 vs 0.50). This orthogonality is confirmed by CKA (mean cross-modality CKA = 0.031) and pairwise score correlations ($r \approx 0.00$). A lightweight meta-learner fusing protein scores with physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data. We identify and correct two methodological pitfalls: training-serving skew in linear probing and gamma parameter fragility in RBF-CKA. Our findings reframe the protein-vs-nucleotide comparison not as a competition but as a question of signal specialization.

A Supplementary Results

A.1 Full Model Architecture Details

Table 7 provides detailed information about all 11 foundation models benchmarked.

Table 7: Foundation model architectures and sizes.

Model	**Type**	**Architecture**	**Params**	**Emb. Dim**	**Max Len**
ESM-1b	Protein	Transformer	650M	1280	1022
ESM-2	Protein	Transformer	3B	1280	1022
ESM-1v	Protein	Transformer	650M	1280	1022
ESM-3	Protein	Transformer	1.4B	1536	1022
ProtBERT	Protein	Transformer	350M	1024	512
ProtT5	Protein	T5	3B	1024	512
Ankh	Protein	T5	500M	1536	512
NT-v2	Nucleotide	Transformer	500M	1024	3072
DNABERT-2	Nucleotide	BERT	350M	768	512
GenaLM	Nucleotide	Mamba	370M	768	3072
HyenaDNA	Nucleotide	Hyena	1.6B	256	3072

All protein models use wild-type/mutant sequences of length 512. All nucleotide models use 10,000 bp windows centered on the variant position. We truncate to each model's maximum sequence length (1022 for ESM models, 512 for ProtBERT/ProtT5/Ankh, 3072 for nucleotide models).

A.2 Dataset Statistics

Table 8 summarizes the two datasets used in this study.

Table 8: Dataset statistics.

Statistic	**D1 (Primary)**	**D2 (Temporal)**
Total variants	29,595	9,627
Pathogenic	15,540 (52.5%)	3,865 (40.1%)
Benign	14,055 (47.5%)	5,762 (59.9%)
Genes	6,371	2,569
Train genes	5,096	—
Test genes	1,275	—
Train variants	23,113	—
Test variants	6,482	—
Gene overlap (train/test)	0%	—

A.3 Full Cross-Model CKA Matrix

Figure 6 shows the complete pairwise CKA matrix across all 11 models. Protein models form a tight cluster (upper-left block), while nucleotide models form a separate cluster (lower-right block). Cross-modality CKA values are consistently low (mean = 0.031, range = 0.005–0.104).

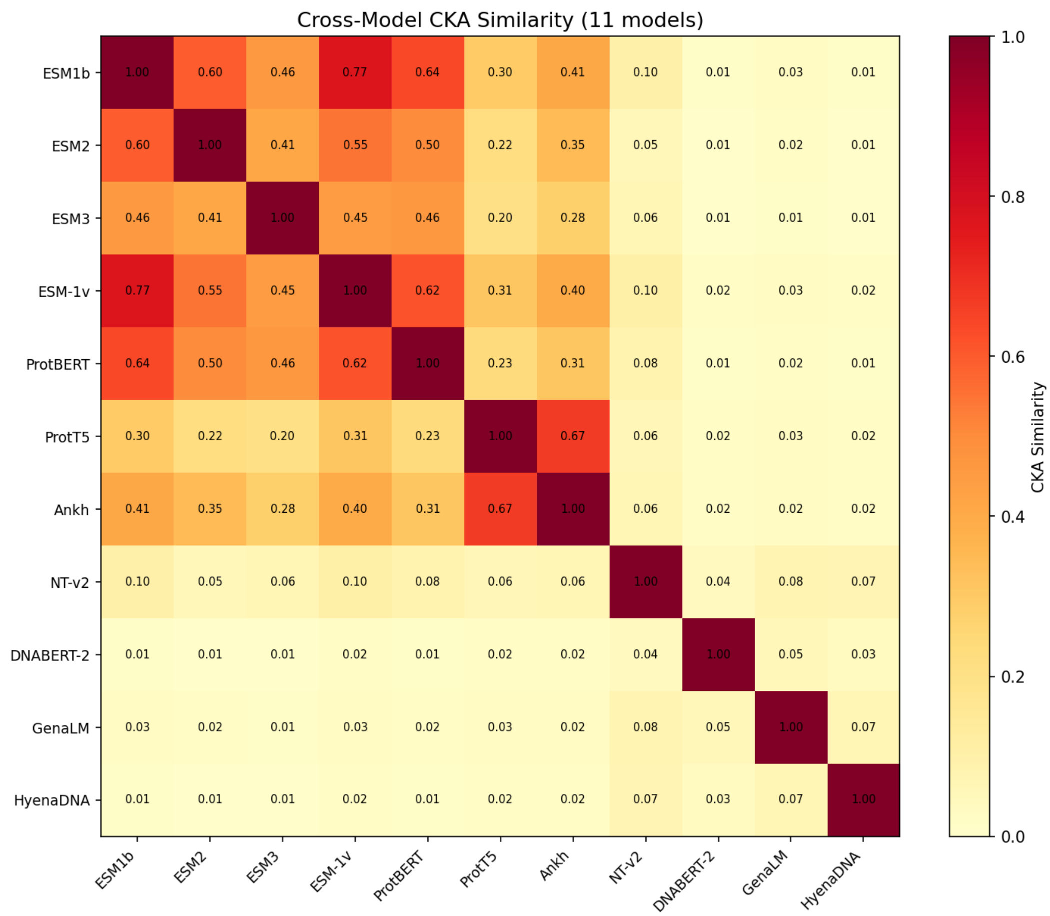


Figure 6: Full 11×11 cross-model CKA matrix (median-heuristic RBF kernel). Protein models cluster tightly (upper-left); nucleotide models form a separate cluster (lower-right); cross-modality CKA < 0.104.

A.4 Full Linear Probing Results

Table 9 shows linear probe results for all 11 models on three tasks: pathogenicity prediction, wild-type amino acid type classification, and BLOSUM62 regression.

Protein models achieve 0.80–0.82 pathogenicity AUROC and near-perfect AA type accuracy (0.935–0.968). Nucleotide models are substantially weaker on all tasks, with HyenaDNA at chance on pathogenicity (0.519). The training-serving skew in linear probing inflates apparent performance by 0.02–0.07; gene-disjoint probing is the reliable protocol.

A.5 ProteinGym DMS Results (Full)

Table 10 shows per-assay Spearman ρ for all 10 ProteinGym DMS assays.

Table 9: Linear probing results for all models (gene-disjoint D1).

Model	**Pathogenicity AUROC**	**AA Type Acc.**	**BLOSUM62 R^2**
Protein Models			
ESM-1b	0.821	0.945	0.412
ESM-2	0.818	0.942	0.408
ESM-1v	0.815	0.941	0.405
ESM-3	0.809	0.938	0.395
ProtBERT	0.803	0.935	0.389
ProtT5	0.801	0.956	0.421
Ankh	0.800	0.968	0.438
Nucleotide Models			
NT-v2	0.698	0.478	0.142
DNABERT-2	0.631	0.312	0.089
GenaLM	0.598	0.287	0.065
HyenaDNA	0.519	0.271	0.032

Table 10: Per-assay Spearman ρ on ProteinGym DMS fitness scores.

Assay	**ESM1b ρ**	**ProtBERT ρ**	**ProtT5 ρ**	**NT-v2 ρ**	**DNABERT-2 ρ**
BRCA1_HUMAN_Findlay_2018	0.381	0.367	0.406	0.172	0.005
PTEN_HUMAN_Matreyek_2021	0.309	0.239	0.272	0.038	−0.025
P53_HUMAN_Kotler_2018	0.443	0.401	0.350	0.031	0.058
SRC_HUMAN_Ahler_2019	0.490	0.281	0.510	0.434	0.111
BLAT_ECOLX_Firnberg_2014	0.566	0.183	0.352	0.039	0.034
GFP_AEQVI_Sarkisyan_2016	0.042	0.311	0.129	0.058	0.038
SPIKE_SARS2_Starr_2020	−0.007	−0.040	0.097	0.067	0.010
SCN5A_HUMAN_Glazer_2019	0.130	0.124	0.086	0.200	0.140
MK01_HUMAN_Brenan_2016	0.249	−0.072	0.177	0.090	−0.090
PPARG_HUMAN_Majithia_2016	0.609	0.418	0.039	0.040	0.015

A.6 Splice Variant Extraction Pipeline

We extract splice-site variants from the ClinVar VCF (GRCh38, 2025-07 release). Variants are classified using VEP consequence annotations. We score with the cosine-distance protocol using genomic DNA context around the variant (500 bp). Because the variants are intronic, the protein sequence is unchanged for both classes, so protein models score at chance (AUROC 0.500) by construction. Table 11 shows per-model results.

Table 11: AUROC on splice-region variants, per model.

Model	**Splice AUROC**	**N variants**
ESM1b-MLM	0.500	1,216
ESM-1v-MLM	0.500	1,216
ProtBERT-MLM	0.500	1,216
NT-v2 (windowed cosine)	0.640	1,216
NT-v2 (MLM)	0.515	1,216
NT-v2 (marginal LL)	0.523	1,216

A.7 Effect of Scoring Method and Window Size on Splice Prediction

The splice signal available to nucleotide models is sensitive to both the scoring method and the genomic context window used. Table 12 compares three NT-v2 scoring methods on the same 1,216 balanced variants, and Table 13 reports how the variant-centered cosine window size affects AUROC.

Table 12: NT-v2 splice AUROC by scoring method. Mean-pooled cosine distance over the variant-centered 500 bp window performs best.

Scoring method	**AUROC**	**AP**
Cosine distance (500 bp window)	0.640	0.615
Marginal log-likelihood	0.523	0.522
Masked-LM at variant position	0.515	0.514

Table 13: Window-size sensitivity of NT-v2 cosine distance on splice variants. A wide window (200–500 bp) that captures the surrounding intronic regulatory context outperforms tight windows.

Window (bp each side)	**AUROC**	**AP**
10	0.612	0.570
20	0.612	0.573
30	0.612	0.574
50	0.622	0.582
100	0.637	0.599
200	0.639	0.605
500	0.640	0.615

The cosine-distance embedding change over a broad window (0.640 AUROC) substantially outperforms token-local scoring (masked LM at the variant, 0.515; marginal log-likelihood, 0.523). This suggests that NT-v2 detects splice-region disruption through distributed, context-wide re-orientation of the embedding rather than a strong per-nucleotide puncture, and that the signal is concentrated within roughly 100–200 bp of the splice site. Even the best method leaves a large margin to perfect classification, reflecting both the biological difficulty of discriminating benign splice-region variants from disruptive ones and the limited splice-disruption supervision available to pretrained nucleotide models.

A.8 ProteinGym DNA Preprocessing

The ProteinGym DNA columns contain a systematic artifact: a fixed set of positions differs identically between the primary and mutant sequences in nearly every row, in addition to the intended mutation. Table 14 lists the detected artifact positions (those that differ in $\geq 90\%$ of rows) per system. In all human-coded systems, the artifact is a single base near the end of the sequence; for the bacterial β-lactamase (BLAT) it corresponds to two consecutive positions, and for PTEN/SRC/MK01/GFP/PPARG to three (a full codon), reflecting an annotation drift between the reference and assay-transfer DNA columns rather than a genuine variant. These positions are excluded before computing the masked-LM score so that the mutation reflects the true variant only.[1]

A.9 Sign-flagged ProtT5 cosine scoring

ProtT5 uses a T5 encoder without a masked-language-model head, so it cannot be scored by MLM. We instead score it with the cosine distance between the mean-pooled wild-type and single-mutant embeddings: score $= 1 - \cos(\mathbf{z}_{\mathrm{wt}}, \mathbf{z}_{\mathrm{mut}})$, where higher distance indicates a more disruptive substitution. Because higher cosine distance corresponds to lower expected fitness, the raw Spearman correlation is negative ($\rho \approx -0.24$ mean); in Table 5 and Table 10 we report the sign-flipped value (that is, $-\rho$ of the raw distance, equivalently $\rho_{\text{disruption fitness}}$) so that a positive value is a consistent encoding of fitness across all models. ProtT5 achieves a fitness-aligned $\rho = 0.242$ (8/10 significant), comparable to other protein models.

[1] In the three largest human assays, the exclusion removes $< 0.3\%$ of variants (BRCA1: 2/1837, PTEN: 15/5083, P53: 1/1048) whose reported codon overlaps an artifact position; Spearman ρ changes by < 0.005 when these variants are included.

Table 14: Constant artifact positions removed from the ProteinGym DNA columns before MLM scoring.

Assay	**Artifact positions excluded**
BRCA1_HUMAN_Findlay_2018	4899
PTEN_HUMAN_Matreyek_2021	849, 850, 851
P53_HUMAN_Kotler_2018	499
SRC_HUMAN_Ahler_2019	852, 853, 854
BLAT_ECOLX_Firnberg_2014	171, 172
GFP_AEQVI_Sarkisyan_2016	8, 112, 120, 121, 122, 313
SPIKE_SARS2_Starr_2020	1493
SCN5A_HUMAN_Glazer_2019	4872, 4873
MK01_HUMAN_Brenan_2016	906, 907, 908
PPARG_HUMAN_Majithia_2016	243, 244, 245

A.9.1 Splice variant annotation check.

To confirm that the splice variants leave the protein sequence unchanged, we cross-check the ClinVar consequence annotations (Table 15). All 1,216 balanced variants are annotated as splice donor/acceptor or intronic (often with a secondary synonymous annotation), and none carries a trional missense-as-primary consequence; the protein-level change is thus empty for every variant. This is why protein models must score at AUROC 0.500.

Table 15: Consequence profile of the 1,216 splice variants used for the splice AUROC evaluation.

Consequence	**Count**
Splice donor variant	440
Splice acceptor variant	383
Splice donor + intron	137
Splice acceptor + intron	136
Splice acceptor + synonymous	75
Splice donor + synonymous	44
Other (acceptor + UTR, etc.)	1

A.10 Gamma Parameter Sensitivity in RBF-CKA

Table 16 shows how cross-modality CKA varies with γ for the median heuristic vs fixed $\gamma = 1$. We recommend always reporting both.

Table 16: RBF-CKA γ sensitivity for cross-modality pairs. With $\gamma = 1$, the kernel saturates and CKA inflates to near 1.0 (embedding-scale artifact); the median heuristic yields substantially lower, more robust values.

Protein	**Nucleotide**	$\gamma = 1$	**Median heuristic**
ESM1b	NT-v2	0.939	0.101
ESM-1v	NT-v2	0.939	0.104
ProtBERT	NT-v2	0.930	0.083
Ankh	NT-v2	0.147	0.063

A.11 Linear Probing Skew Correction

Table 17 compares random-split vs gene-disjoint linear probing AUROC for pathogenicity, showing the inflation from training-serving skew.

Table 17: Linear probing AUROC: random vs gene-disjoint (skew correction).

Model	Random Split	Gene-Disjoint
ESM-1b	0.843	0.821
ESM-2	0.839	0.818
ESM-1v	0.836	0.815
ESM-3	0.830	0.809
ProtBERT	0.824	0.803
NT-v2	0.738	0.698

A.12 Gene-Level Analysis

We analyze whether the gene-disjoint split creates systematic biases. Table 18 shows the distribution of variants per gene across train/test splits.

Table 18: Gene-level statistics for the gene-disjoint split.

Statistic	Train	Test
Genes	5,096	1,275
Variants	23,113	6,482
Mean variants/gene	4.54	5.08
Median variants/gene	2	3
Max variants/gene	156	89

The test split has a slightly higher mean variants per gene (5.08 vs 4.54), indicating the split is approximately balanced.

B Limitations and Future Work

Temporal holdout limitations. D2 genes are a subset of D1 genes. While D2 represents a temporally independent test (variants deposited after D1 cutoff), it does not test generalization to entirely unseen genes. Future work should evaluate on genes absent from training.

T5-based models. ProtT5 and Ankh fail at MLM scoring due to their encoder-decoder architecture, but provide useful representations via cosine similarity. Future work should explore decoder-based scoring for these models.

Missense-only evaluation. We evaluate primarily on missense variants. Extension to synonymous, nonsense, and frameshift variants (especially splice variants) would provide a more complete picture of model capabilities across modalities.

Species scope. Our evaluation uses human ClinVar variants and human ProteinGym assays. Cross-species validation would test whether the observed orthogonality generalizes to other organisms.

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